CEE6501 — Lecture 11.1

Introduction to Nonlinearity

Learning Objectives

By the end of this lecture, you will be able to:

- explain the difference between **linear** and **nonlinear** structural analysis
- distinguish between **material** and **geometric** nonlinearity
- describe the main **levels of analysis** used in structural engineering
- explain the course focus on **geometric nonlinearity** for **axial-member structures**
- explain why a simple plane truss can have a **nonlinear load-displacement response** even with linearly elastic material
- describe the ideas of **critical point** and **snap-through instability**

Agenda

- Part 1 — What Is Nonlinearity?
- Part 2 — Types of Nonlinearity
- Part 3 — Levels of Analysis
- Part 4 — Scope for This Course
- Part 5 — Basic Concept of Geometrically Nonlinear Analysis
- Part 6 — Example of Linear vs. Geometrically Nonlinear Analysis

Part 1 — What Is Nonlinearity?

> *We begin by stepping beyond the assumptions of ordinary linear structural analysis.*

Review: What Linear Analysis Assumes

In the usual **linear** structural analysis, we rely on two key assumptions:

- **Geometric linearity**: deformations are small enough that equilibrium can be written on the **undeformed geometry**
- **Material linearity**: stress is proportional to strain, so the material remains **linearly elastic**

In its simplest form, linear structural analysis gives a relation such as

$$\mathbf{K}\mathbf{u} = \mathbf{f}$$

where:

- $\mathbf{K}$ is constant
- response is proportional to load
- superposition holds

This is often adequate for many structures under ordinary **service-load** conditions.

A Simple Mathematical View

In a **linear** structural problem, the applied load and the response are proportional.

For a single degree of freedom spring:

$$f = ku$$

where:

- f = applied force
- u = displacement
- k = stiffness

If k is constant, then doubling the load doubles the displacement:

$$2f = k \times 2u$$

So the force-displacement curve is a **straight line**.

In a **nonlinear** problem, that proportional relationship no longer holds.

The response may instead look like.

$$f = f(u)$$

which means the force, f, is some nonlinear function of displacement, u.

For example, the slope of a member may change as the structure deforms, so the stiffness is no longer constant:

$$k_{\text{tangent}} = \frac{df}{du}$$

Why Nonlinear Analysis Is Needed

As loads increase, or as deformations become significant, the assumptions of linear analysis begin to break down.

A structure may become nonlinear because:

- its **geometry changes enough** during deformation to affect equilibrium and internal force
- its **material behavior changes**, for example through yielding, cracking, or inelastic response

Then the structural response is no longer a straight-line relation between load and displacement.

Main idea: in a nonlinear problem, the structure you are solving for also changes while you solve it.

Nonlinear Equilibrium in the Structural Context

At the structural level, we usually write the problem in terms of **equilibrium** between external and internal forces:

$$\mathbf{r}(\mathbf{u}) = \mathbf{f}_{\text{ext}} - \mathbf{f}_{\text{int}}(\mathbf{u}) = \mathbf{0}$$

Here:

- $\mathbf{f}_{\text{ext}}$ is the applied load vector
- $\mathbf{f}_{\text{int}}(\mathbf{u})$ is the internal resisting force vector, written as a function of the current nodal position
- $\mathbf{r}(\mathbf{u})$ is the **residual**, or out-of-balance force vector, also written as a function of the current nodal position

So in a nonlinear analysis, the goal is to find the displacement state $\mathbf{u}$ for which the structure is in equilibrium.

This is harder than the linear case because:

- the internal force depends on the current displacement
- the stiffness may change as the structure deforms
- the solution usually must be found **iteratively**

Why This Matters in Structural Engineering

Linear analysis is often very useful, but it has inherent limitations.

It may not capture:

- loss of stiffness caused by large deformations
- second-order effects such as P-Δ and P-δ
- instability phenomena such as buckling
- inelastic material behavior and strength degradation

So nonlinear analysis becomes important when the goal is not only to compute forces and displacements under service conditions, but also to understand:

- ultimate strength
- stability
- large-deformation behavior
- failure mechanisms

Part 2 — Types of Nonlinearity

Material Nonlinearity

One major source of structural nonlinearity is **material nonlinearity**.

The constitutive law is nonlinear, so the relationship between stress and strain is no longer a straight line.

For a linear elastic material, we write

$$\sigma = E\varepsilon$$

where the slope E is constant.

For a nonlinear material, the stress is some more general function of strain:

$$\sigma = \sigma(\varepsilon)$$

So the material stiffness is no longer constant. Instead, the slope changes as the material deforms:

$$E_{tangent} = \frac{d\sigma}{d\varepsilon}$$

Examples of Material Nonlinearity

Material nonlinearity arises when the stress-strain law is not linear elastic.

Typical examples include:

- yielding of steel
- cracking of reinforced concrete
- creep or time-dependent deformation
- nonlinear interaction of axial force, bending, shear, and torsion

Conceptually:

- the material stiffness changes as loading proceeds
- unloading and reloading may follow different paths
- the load-displacement curve can become history-dependent

For a linear elastic material, the slope is constant. For a nonlinear material, the tangent stiffness changes along the response path.

Geometric Nonlinearity

The second major source is **geometric nonlinearity**.

The deformation is large enough that the changing geometry affects equilibrium, compatibility, and internal force.

In a linear analysis, we treat the structure as if its geometry does not change significantly during loading, so the stiffness is taken as fixed.

In a geometrically nonlinear analysis, the current deformed shape affects the response, so the stiffness depends on the structure's current nodal positions:

$$\mathbf{K} = \mathbf{K}(\mathbf{u})$$

Equivalently, the internal resisting forces depend on the current nodal positions:

$$\mathbf{f}_{\text{int}} = \mathbf{f}_{\text{int}}(\mathbf{u})$$

Examples of Geometric Nonlinearity

Geometric nonlinearity arises when deformations become large enough that the current geometry cannot be ignored.

Typical examples include:

- P-Δ effect: gravity load acting through a lateral story displacement
- P-δ effect: axial force interacting with member curvature
- cables, membranes, or very flexible trusses whose stiffness depends strongly on shape (need to be form-found)
- structures undergoing large displacement before reaching material failure

Conceptually:

- equilibrium must be written on the **deformed** configuration
- internal force depends on both displacement and applied load
- stiffness becomes configuration-dependent

Combined Nonlinear Effects

In many realistic structural problems, the two types of nonlinearity occur together.

Examples include:

- a steel frame with both **yielding** and P**-**Δ **effects**
- reinforced concrete members with **cracking** and **large deflections**
- frames where connection deformation, panel-zone deformation, and stability effects all interact

A useful way to think about it is:

- **material nonlinearity** changes the constitutive law
- **geometric nonlinearity** changes the equilibrium problem

In a first introduction, it is common to isolate one source of nonlinearity at a time.

Part 3 — Levels of Analysis

Why We Talk About Levels of Analysis

It is rarely practical to model every source of nonlinearity in full detail.

So structural analysis is often organized into **levels of analysis**, where each level captures more of the true behavior than the one before it.

These levels differ in:

- what assumptions are retained
- what sources of nonlinearity are included
- what kinds of failure or instability they can detect
- the computational effort required

Each higher level is not just "more complicated" — it is answering a different question about the structure.

First-Order Elastic Analysis

This is the standard **linear** analysis we have used so far.

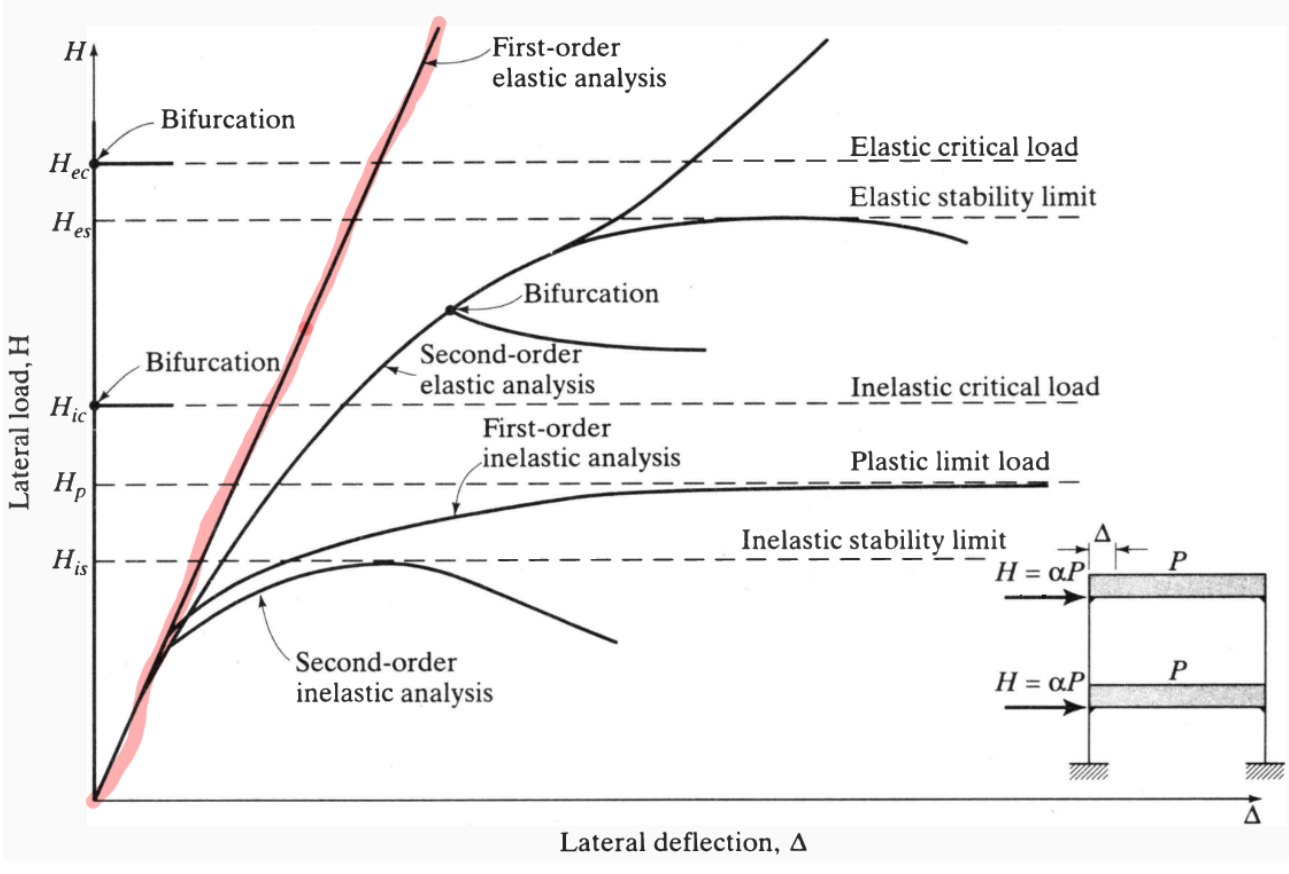
Assumptions

- equilibrium written on the **undeformed** structure
- material remains **linearly elastic**
- stiffness remains constant

What it misses

- geometric nonlinear effects
- material nonlinear effects
- instability behavior

$$\mathbf{K}\mathbf{u} = \mathbf{f}$$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Lateral load, H
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Elastic critical load
Elastic stability limit
Bifurcation
Second-order
elastic analysis
Inelastic critical load
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Plastic limit load
Inelastic stability limit
Second-order
inelastic analysis
Δ
$H = \alpha P$
P
$H = \alpha P$
P
Δ
Lateral deflection, Δ

Second-Order Elastic Analysis

This level includes the effects of deformation on equilibrium while still assuming elastic material behavior.

So it can represent:

- finite displacement effects
- P-Δ effects
- P-δ effects
- some aspects of stability and load amplification

But it still does **not** represent yielding, cracking, or other material nonlinearity.

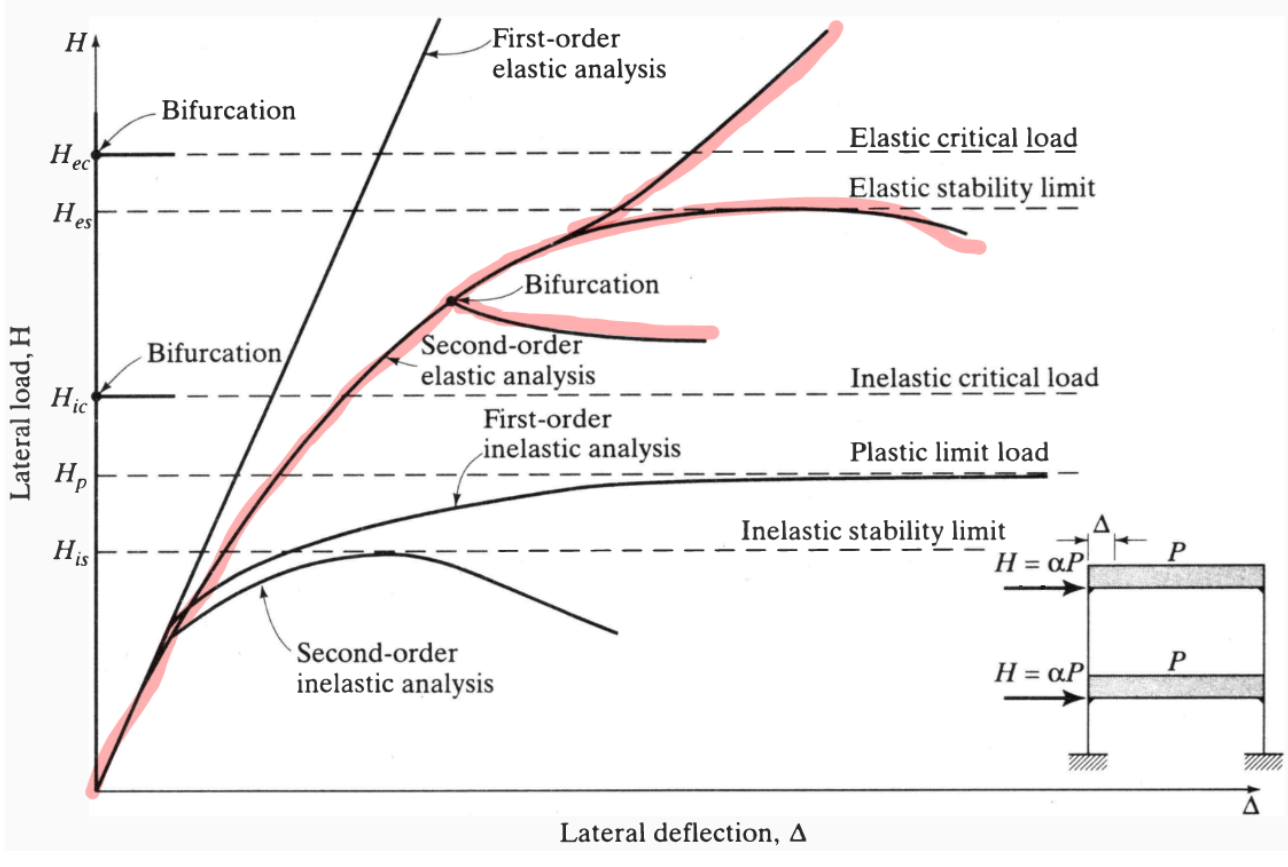
Symbolically, the problem begins to look like

$$\mathbf{K}(\mathbf{u})\mathbf{u} = \mathbf{f}$$

or, more carefully, as a sequence of updated equilibrium states.

Several possible modes of nonlinear elastic behavior are shown.

- **bifurcation** of the load path into an alternative branch
- gradual softening that culminates in an **elastic instability** at a limit point
- stiffness changes that may lead to either softening or stiffening behavior



First-Order Inelastic and Second-Order Inelastic Analysis

More advanced levels of analysis include material nonlinearity, with or without geometric nonlinearity.

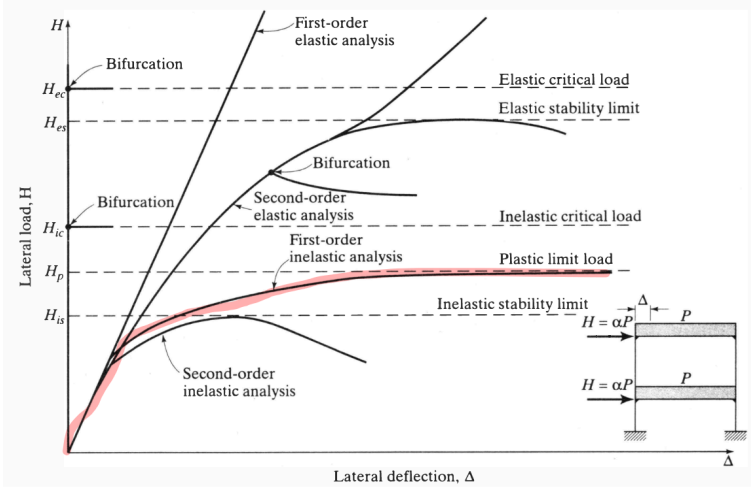
First-order inelastic

- includes material nonlinearity
- keeps equilibrium on the undeformed geometry
- can represent plastic response, but not geometric nonlinear instability effects

Second-order inelastic

- includes both **material** and **geometric** nonlinearity
- is the most general of these common levels
- can represent strength loss, stability limits, and strongly nonlinear response

Much more involved than what we need for our immediate goals.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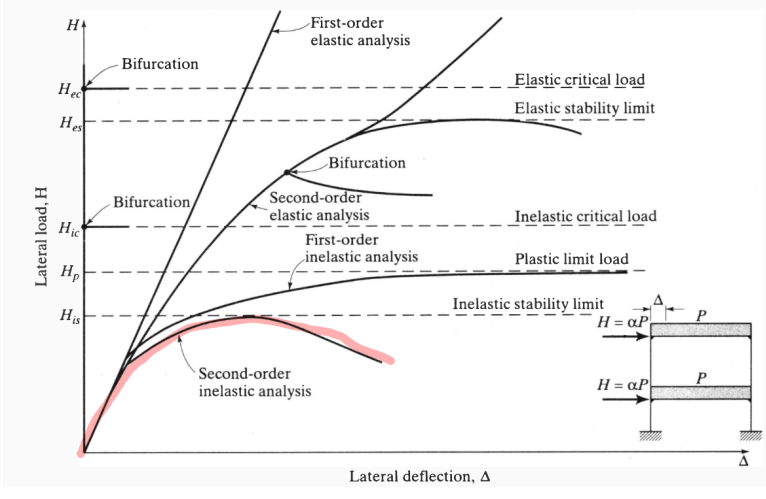
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Summary of the Levels of Analysis

Level of analysis	Geometric nonlinearity?	Material nonlinearity?	Typical use
First-order elastic	No	No	Basic service-load response
Second-order elastic	Yes	No	Large-displacement elastic response, stability effects
First-order inelastic	No	Yes	Inelastic material response without geometric effects
Second-order inelastic	Yes	Yes	Most general response including strength and stability

Part 4 — Scope for This Course

What We Will Focus On

In CEE6501, we will **not** attempt a general treatment of all nonlinear structural analysis.

Instead, we will focus on:

- geometric nonlinearity
- axial elements only
- bar and joint frameworks (trusses, space frames, cable nets, etc.)
- initially in 2D plane trusses

So the goal is not a full nonlinear analysis of general frames or continuous surfaces.

The goal is to understand the **basic concepts of geometrically nonlinear analysis** for axial-member structures whose stiffness and equilibrium depend strongly on their deformed shape.

Why This Narrower Scope Makes Sense

This narrower focus connects directly to **form finding**, which is fundamentally an inverse problem:

- instead of asking for the response of a known structure
- we ask what **structural form** results from a given set of inputs, constraints, and force patterns

That requires the structure to **change shape** until forces come into equilibrium, so it is naturally formulated as a **geometrically nonlinear** problem.

That is exactly the setting for methods such as:

- force density methods (next lecture)
- dynamic relaxation (last lecture)

So before we can find shapes that are in equilibrium, we first need a basic understanding of **geometric nonlinearity**.

Textbook Readings

McGuire provides the broader background for nonlinear analysis, especially in:

- **Chapter 8** — introduction to nonlinear behavior and analysis
- **Chapter 9** — geometric nonlinearity and elastic critical load analysis
- **Chapter 10** — material nonlinear analysis

From McGuire, we take the big-picture framework:

- what nonlinearity is
- the main types of nonlinearity
- the common levels of analysis

From here, we narrow the scope.

Following Kassimali (Chapter 10) more closely, we will study only the **basic concepts of geometrically nonlinear analysis** for **plane trusses**.

Our focus will be on:

- axial elements only
- large displacements
- equilibrium on the deformed geometry
- iterative solution and convergence

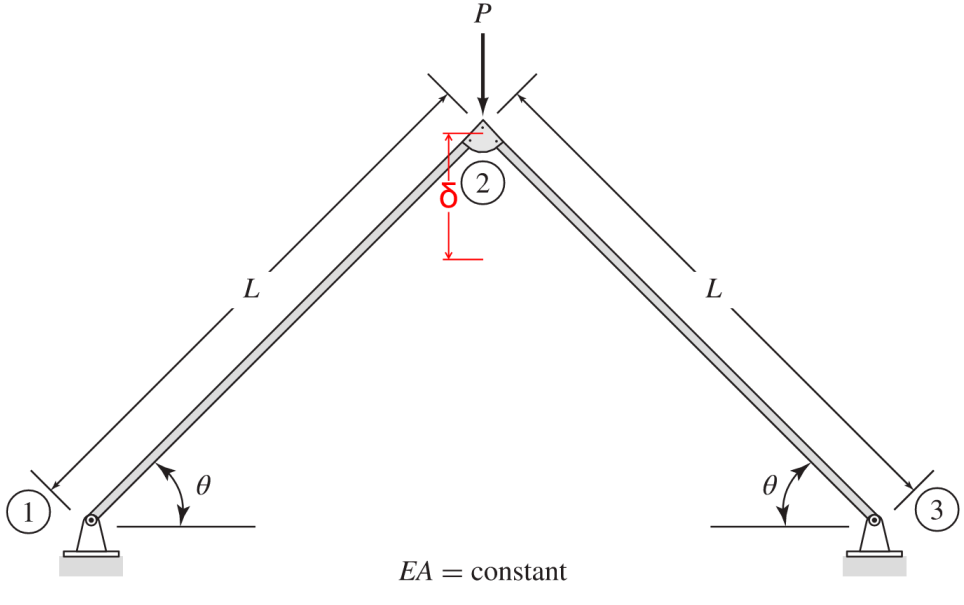
Part 5 — Basic Concept of Geometrically Nonlinear Analysis

The Example Structure

To illustrate the basic concept of **geometrically nonlinear analysis**, we consider a simple symmetric **two-member plane truss** made of a linearly elastic material.

The structure is loaded by a vertical load P at the top joint, so the motion is described by a single degree of freedom:

$$\delta = \text{vertical displacement of joint } 2$$



Because the structure is symmetric:

- both members have the same undeformed length L
- both members have the same area A
- both members have the same angle θ in the undeformed configuration
- both members carry the same axial force magnitude Q

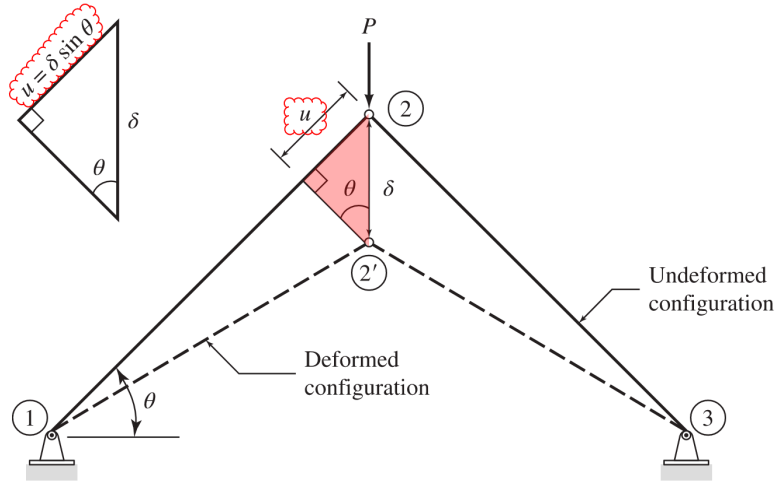
This very simple system is enough to show the essential difference between **linear** and **geometrically nonlinear** analysis.

Linear Analysis: Small-Displacement Kinematics

In **linear analysis**, the displacement δ is assumed to be small enough that the member axial deformation can be approximated from the **undeformed** geometry.

For each member,

$$u \approx \delta \sin \theta \tag{1}$$



so the axial strain is

$$\varepsilon = \frac{u}{L} \approx \left(\frac{\sin\theta}{L}\right)\delta \tag{2}$$

This is a **linear** relation between strain and joint displacement.

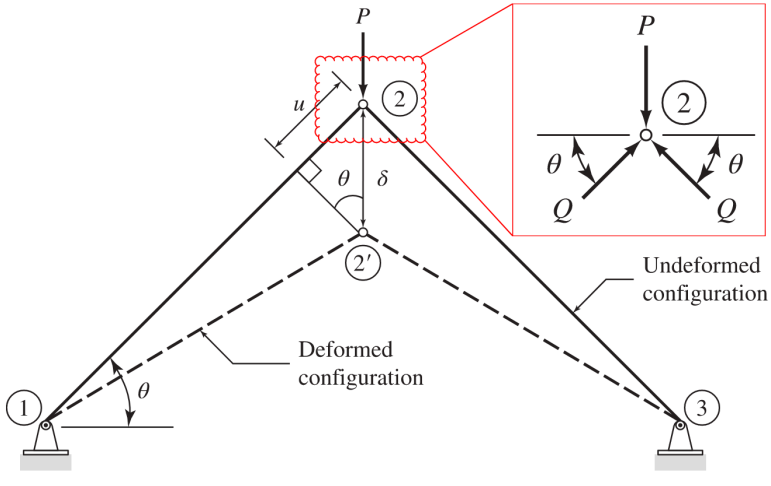
The key approximation is that the member direction is taken from the **original configuration**, not the deformed one ($L = L_0$)

Linear Analysis: Equilibrium on the Undeformed Geometry

In linear analysis, the equilibrium equations are also written on the **undeformed** geometry.

At the loaded top joint, vertical equilibrium gives

$$Q \approx \frac{P}{2\sin\theta} \tag{3}$$



and therefore the axial stress is

$$\sigma = \frac{Q}{A} \approx \frac{P}{2A\sin\theta} \tag{4}$$

For a linearly elastic material,

$$\sigma = E\varepsilon \tag{5}$$

Combining the linear expressions for stress, Eq. (4), and strain, Eq. (2), gives the linear load-displacement relation:

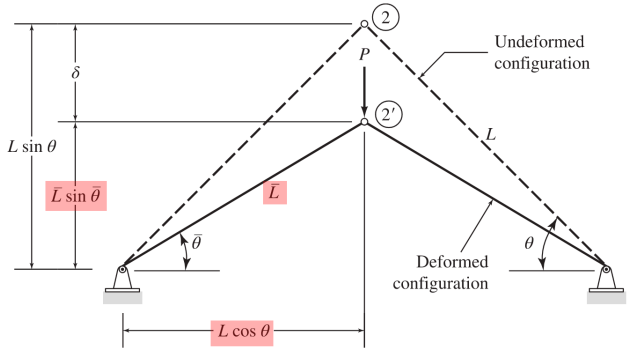
$$P \approx \underbrace{\left(\frac{2EA\sin^2\theta}{L}\right)}_{\text{stiffness term}}\delta \tag{6}$$

So in the linear model, the structure behaves like a spring with constant stiffness:

$$k = \frac{2EA\sin^2\theta}{L}$$

Geometrically Nonlinear Analysis: Use the Deformed Geometry

We do not assume the displacement is small.



$$\bar{L} = \sqrt{(L\cos\theta)^2 + (L\sin\theta - \delta)^2} = L\sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta} \tag{7}$$

$$\sin\bar{\theta} = \frac{L\sin\theta - \delta}{\bar{L}} = \frac{\sin\theta - \left(\frac{\delta}{L}\right)}{\sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta}} \tag{8}$$

Exact Axial Deformation and Strain

The **first key difference** from linear analysis is in the **kinematics**. The member deformation is computed from the **actual deformed length**.

The exact axial deformation is therefore

$$u = \bar{L} - L$$

Substituting the expression for the deformed member length from Eq. (7) gives

$$u = L\left[\sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta} - 1\right] \tag{9}$$

and the corresponding axial strain becomes

$$\varepsilon = \frac{u}{L} = \sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta} - 1 \tag{10}$$

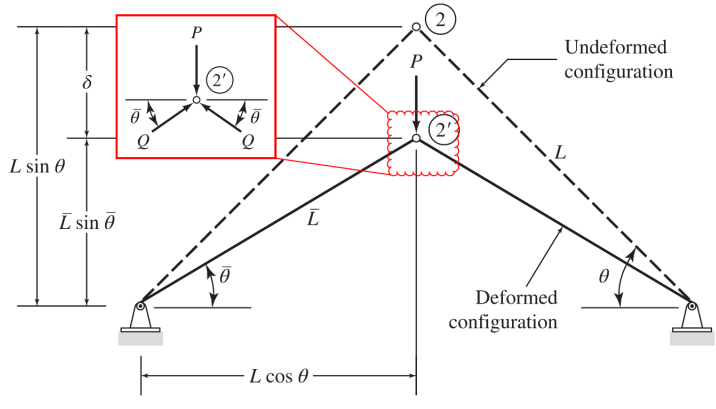
So, unlike the linear case, both axial deformation and strain are now **nonlinear functions** of the joint displacement δ.

Equilibrium Must Also Be Written on the Deformed Geometry

The **second** key difference from linear analysis is in the equilibrium equations since when the current member angle changes, so too does the vertical equilibrium relation.

vertical equilibrium at the top joint

$$Q = \frac{P}{2\sin\bar{\theta}} \tag{11}$$



So unlike the linear case, the internal member force is not related to P through the original angle θ, but through the **current** angle $\bar{\theta}$.

Substituting the expression for the deformed angle from Eq. (8) gives

$$Q = \frac{P\sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta}}{2\left[\sin\theta - \left(\frac{\delta}{L}\right)\right]}$$

which yields the expression for member axial stress as

$$\sigma = \frac{Q}{A} = \frac{P\sqrt{1 + \left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta}}{2A\left[\sin\theta - \left(\frac{\delta}{L}\right)\right]} \tag{12}$$

Finally, by substituting the expressions for strain, Eq. (10), and stress, Eq. (12), into the linear stress-strain relationship $\sigma = E\varepsilon$, we obtain the nonlinear load-displacement relation for the truss:

$$P = 2EA\left[\sin\theta - \left(\frac{\delta}{L}\right)\right]\left[\frac{1-\sqrt{1+\left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta}}{\sqrt{1+\left(\frac{\delta}{L}\right)^2 - 2\left(\frac{\delta}{L}\right)\sin\theta}}\right] \tag{13}$$

Result: A Nonlinear Load-Displacement Relation

After combining:

- the exact deformed geometry
- the exact strain expression
- Hooke's law, $\sigma = E\varepsilon$
- and equilibrium written on the deformed configuration

we obtain a **nonlinear** relation between load P and displacement δ.

The key point is that the load is now a function of displacement:

$$P = P(\delta)$$

rather than a linear relation of the form

$$P = k\delta$$

with constant stiffness k.

So even if the material remains **linearly elastic**, the overall structural response can still be **nonlinear**, because the geometry changes as the structure deforms.

Summary — Two Key Differences

Kinematics

In linear analysis,

$$u \approx \delta \sin \theta$$

In geometrically nonlinear analysis,

$$u = \bar{L} - L$$

So the member deformation is based on the **exact deformed geometry**, not a small-displacement approximation.

Equilibrium

In linear analysis, equilibrium is written on the **undeformed** geometry.

In geometrically nonlinear analysis, equilibrium is written on the **deformed** geometry.

So both the deformation and the equilibrium equations depend on the **current shape** of the structure.

Part 6 - Example of Linear vs. Geometrically Nonlinear Analysis

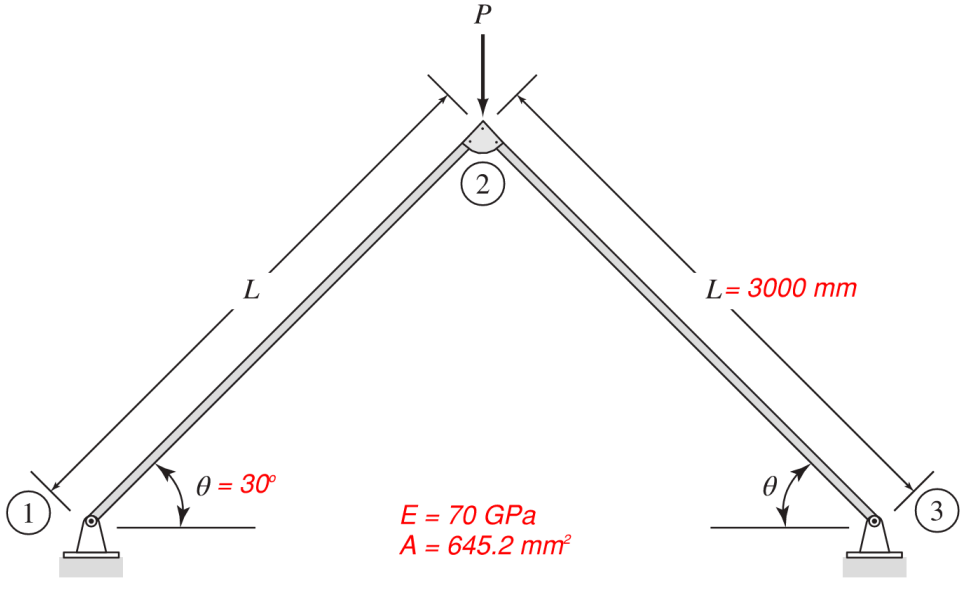
Example Setup

Consider the shallow two-member truss shown below.

We will compare:

- the **linear** load-displacement prediction
- the **geometrically nonlinear** load-displacement prediction

for the same structure and material properties.



In [1]:

```python
import numpy as np
import matplotlib.pyplot as plt

# Two-bar truss properties
E = 70e3              # MPa = N/mm^2
A = 645.2             # mm^2
L = 3000              # mm
theta = np.deg2rad(30)

print(f"E = {E:.0f} MPa")
print(f"A = {A:.1f} mm^2")
print(f"L = {L:.0f} mm")
print(f"theta = {np.rad2deg(theta):.1f} deg")
```

```
E = 70000 MPa
A = 645.2 mm^2
L = 3000 mm
theta = 30.0 deg
```

In [2]:
```
delta = np.arange(0, 3500, 5)  # mm

# Linear model
P_lin = (2 * E * A * np.sin(theta)**2 / L) * delta

# Geometrically nonlinear model from the exact truss geometry
x = delta / L
root = np.sqrt(1 + x**2 - 2*x*np.sin(theta))
P_geo = 2 * E * A * (np.sin(theta) - x) * (1 - root) / root
```

Equations Used in the Example

The **linear** model is

$$P_{\text{lin}} = \left(\frac{2EA \sin^2 \theta}{L} \right) \delta \tag{6}$$

The **geometrically nonlinear** model is based on the exact relation

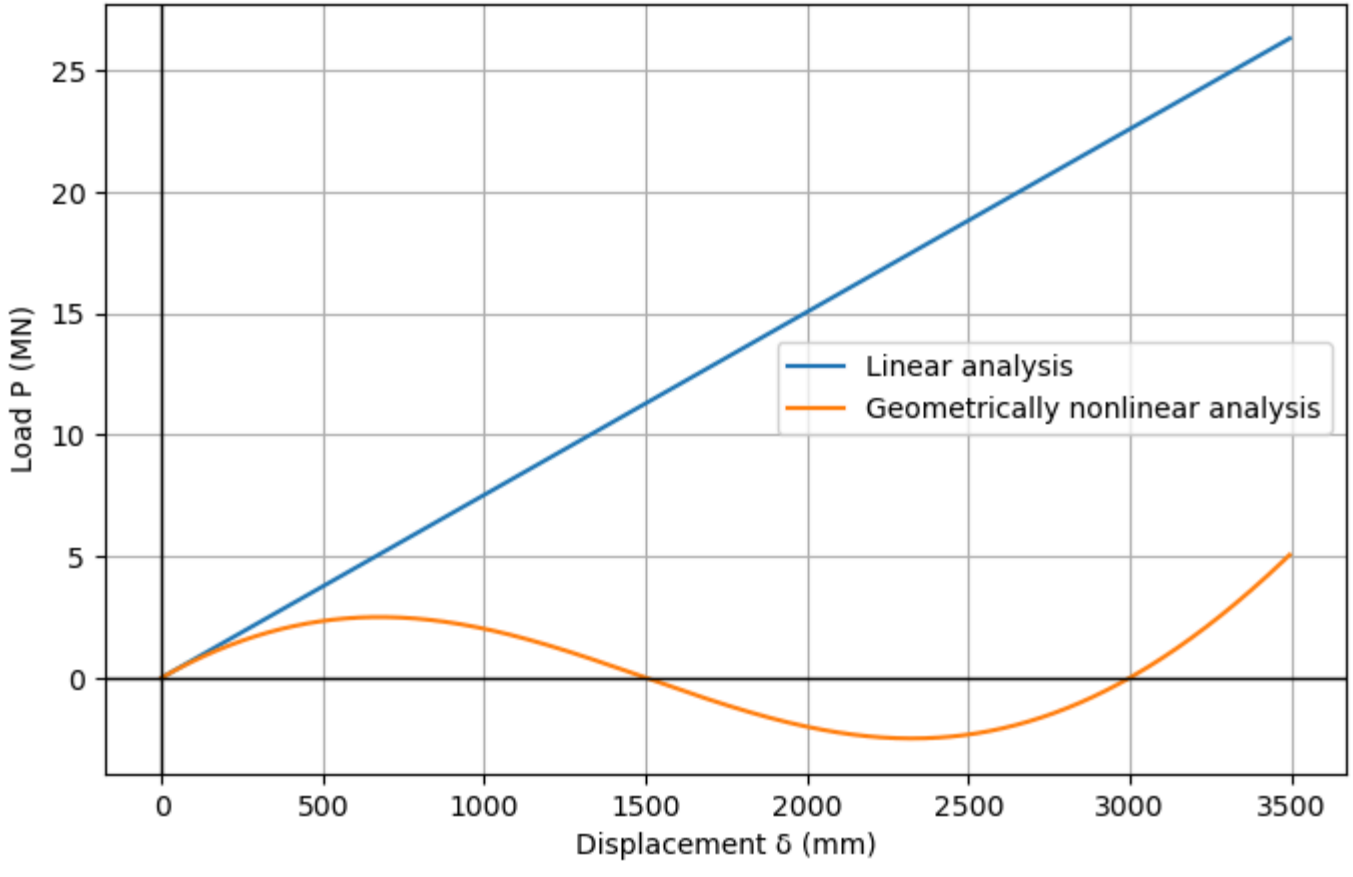
$$P = 2EA \left[\sin\theta - \left(\frac{\delta}{L} \right) \right] \left[\frac{1 - \sqrt{1 + \left(\frac{\delta}{L} \right)^2 - 2 \left(\frac{\delta}{L} \right) \sin\theta}}{\sqrt{1 + \left(\frac{\delta}{L} \right)^2 - 2 \left(\frac{\delta}{L} \right) \sin\theta}} \right] \tag{13}$$

In [3]:
```
def plot_response(delta, P_lin, P_geo):
    plt.figure(figsize=(8, 5))
    plt.plot(delta, P_lin / 1e6, label='Linear analysis')
    plt.plot(delta, P_geo / 1e6, label='Geometrically nonlinear analysis')

    plt.axhline(0, color='black', linewidth=1)
    plt.axvline(0, color='black', linewidth=1)

    plt.xlabel('Displacement δ (mm)')
    plt.ylabel('Load P (MN)')
    plt.legend()
    plt.grid(True)
    plt.show()
```

In [4]:
```
plot_response(delta, P_lin, P_geo)
```



Interpreting the Response Curve

From the plot, we observe:

- the **linear** solution increases indefinitely with constant slope
- the **geometrically nonlinear** solution gradually softens as displacement increases
- the two predictions agree at small displacement, then diverge more and more as deformation grows

This means the linear approximation becomes progressively less accurate as the geometry change becomes significant.

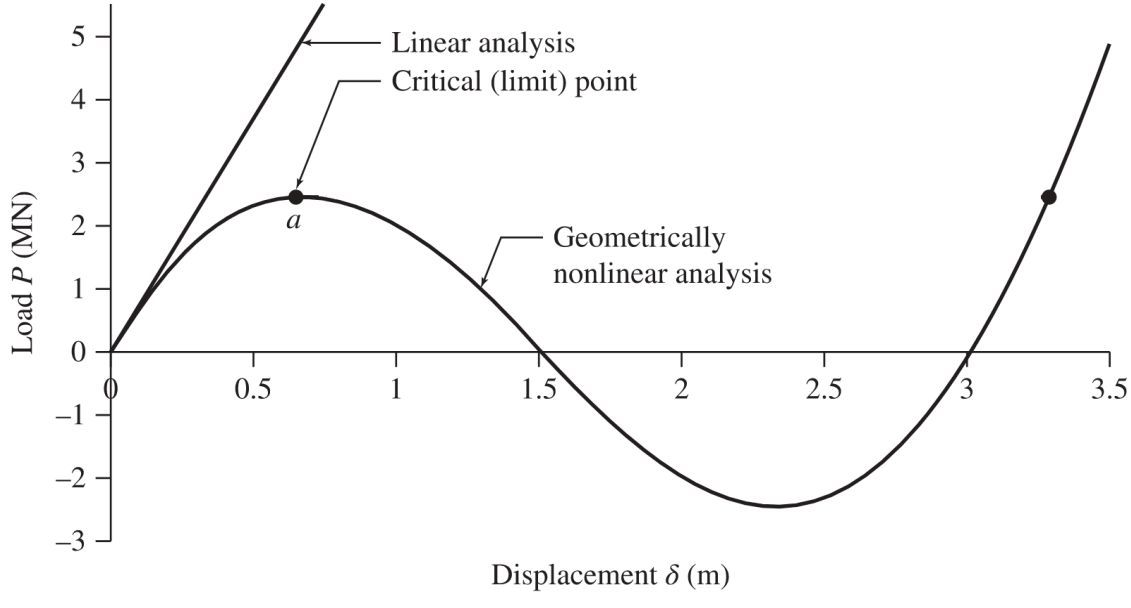
Critical Point and Tangent Stiffness

The slope of the nonlinear response curve is the **tangent stiffness**.

At its peak the tangent stiffness becomes zero:

$$\frac{dP}{d\delta} = 0$$

When it reaches 0, the structure has reached a **critical** or **limit point**.



For the shallow two-member truss, the corresponding critical displacement is

$$\delta_{cr} = L\left[\sin\theta - \cos\theta\sqrt{\frac{1}{(\cos\theta)^{2/3}} - 1}\right] \tag{14}$$

Substituting δ_{cr} into Eq. (13) gives the critical load P_{cr}.

This is the point where the structure loses its ability to resist any further increase in load along the same equilibrium path.

In [5]:
```
delta_cr = L * (np.sin(theta) - np.cos(theta) * np.sqrt(1 / (np.cos(theta)**(2

x_cr = delta_cr / L
root_cr = np.sqrt(1 + x_cr**2 - 2 * x_cr * np.sin(theta))
P_cr = 2 * E * A * (np.sin(theta) - x_cr) * (1 - root_cr) / root_cr

print(f"Critical displacement δ_cr = {delta_cr:.2f} mm")
print(f"Critical load P_cr = {P_cr / 1e6:.3f} MN")
```

```
Critical displacement δ_cr = 675.78 mm
Critical load P_cr = 2.498 MN
```

Physical Meaning of the Critical Point

The critical point marks the end of stable load increase along the original equilibrium branch.

Before that point:

- additional displacement requires additional load
- the structure still has positive tangent stiffness

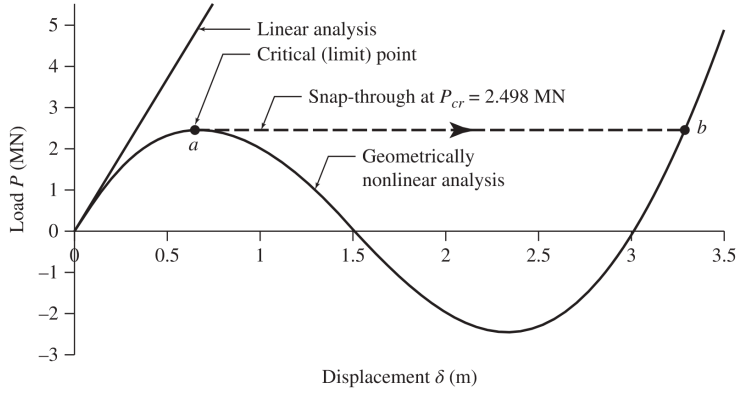
At the critical point:

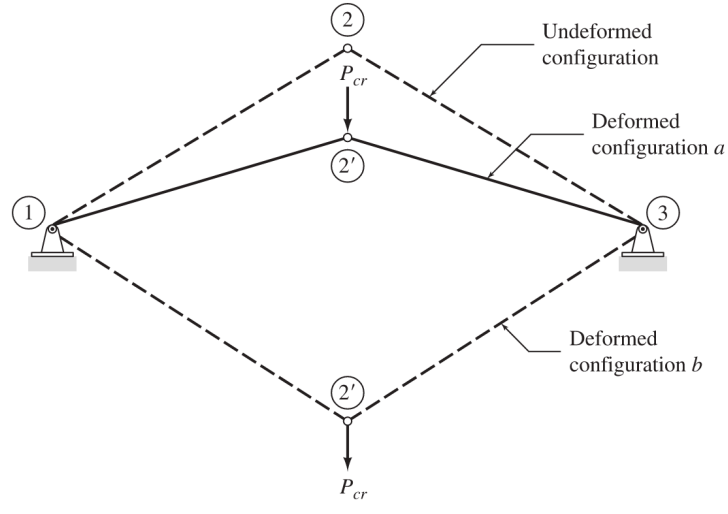
- the tangent stiffness becomes zero
- the response curve reaches a peak
- the structure is on the verge of instability

Snap-Through Instability

Beyond the critical point, the truss cannot continue to carry additional load along the same equilibrium path.

Instead, it may undergo **snap-through**, in which a small change in loading triggers a sudden jump to a very different configuration.





Why Snap-Through Matters

This behavior is important because:

- **large changes in shape** can occur suddenly
- the structure can lose stiffness before any material failure occurs
- the **linear model gives no warning** that this instability is approaching

The two-member truss, often called the **von Mises truss**, is also important historically and computationally.

Because its behavior is simple enough to study carefully, but rich enough to show true geometric instability, it is commonly used as a **benchmark problem** for validating geometrically nonlinear analysis programs.

Why This Example Matters

This example captures the central idea of **geometric nonlinearity**:

- the material may remain **linearly elastic**
- but the **structural response** can still become strongly nonlinear
- the source of that nonlinearity is the **changing geometry** of the structure

That is exactly why geometric nonlinear analysis matters for:

- flexible structures with large deformations
- stability problems
- equilibrium shape finding
- later form-finding methods such as **force density** and **dynamic relaxation**

Wrap-Up

In this lecture, we:

- introduced the basic idea of **structural nonlinearity**
- distinguished between **material** and **geometric** nonlinearity
- reviewed the main **levels of analysis**
- defined the course focus on **geometrically nonlinear axial-member structures**
- used a simple two-member truss to show how **changing geometry** leads to a nonlinear load-displacement response
- introduced the ideas of **critical point** and **snap-through instability**

Next: we will begin developing the member-level equations for **geometrically nonlinear plane truss analysis** and implement a basic **Newton-Raphson nonlinear solver**.